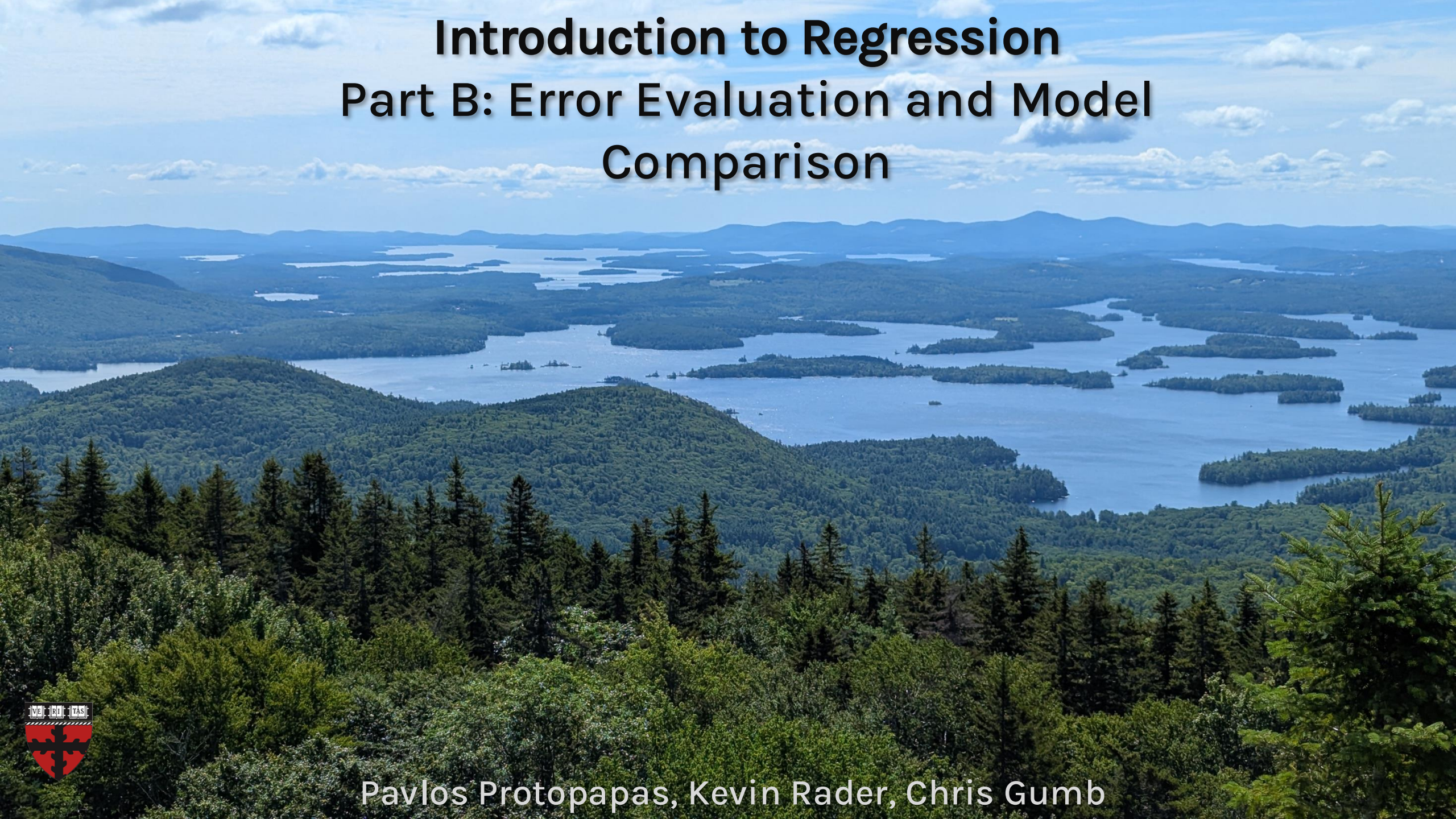
Introduction to Regression

Part B: Error Evaluation and Model Comparison



Pavlos Protopapas, Kevin Rader, Chris Gumb

Lecture Outline

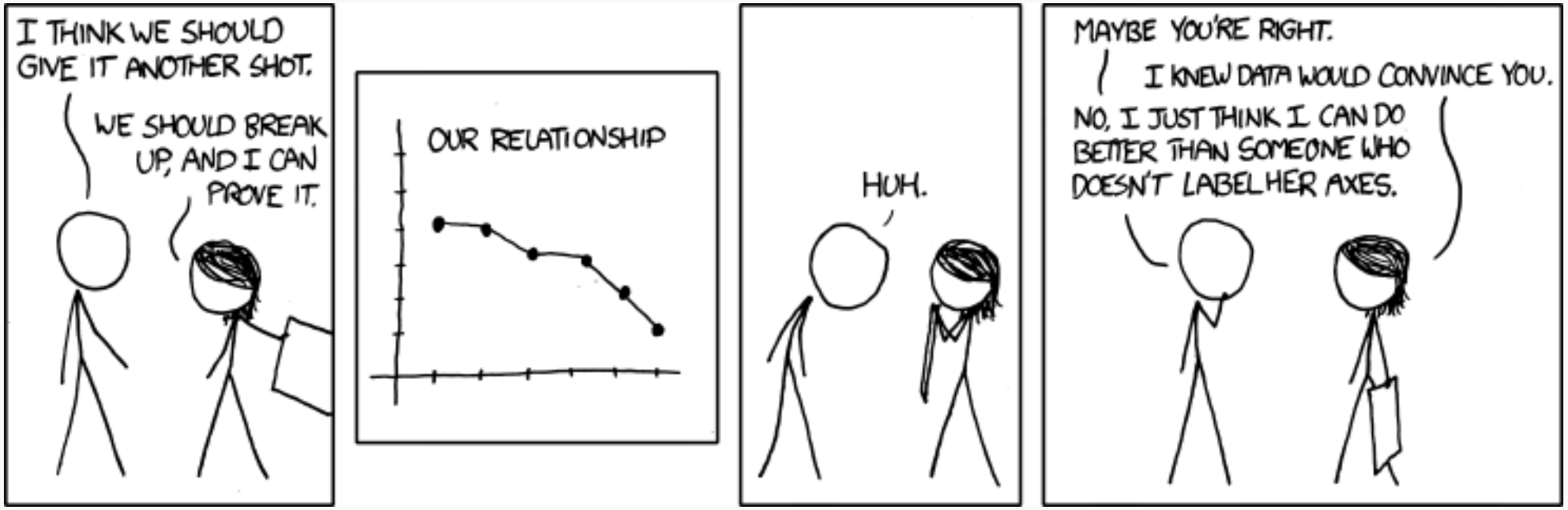
Part A: Statistical Modeling

k-Nearest Neighbors (kNN)

Part B: Error Evaluation and Model Comparison

How do we evaluate our model?

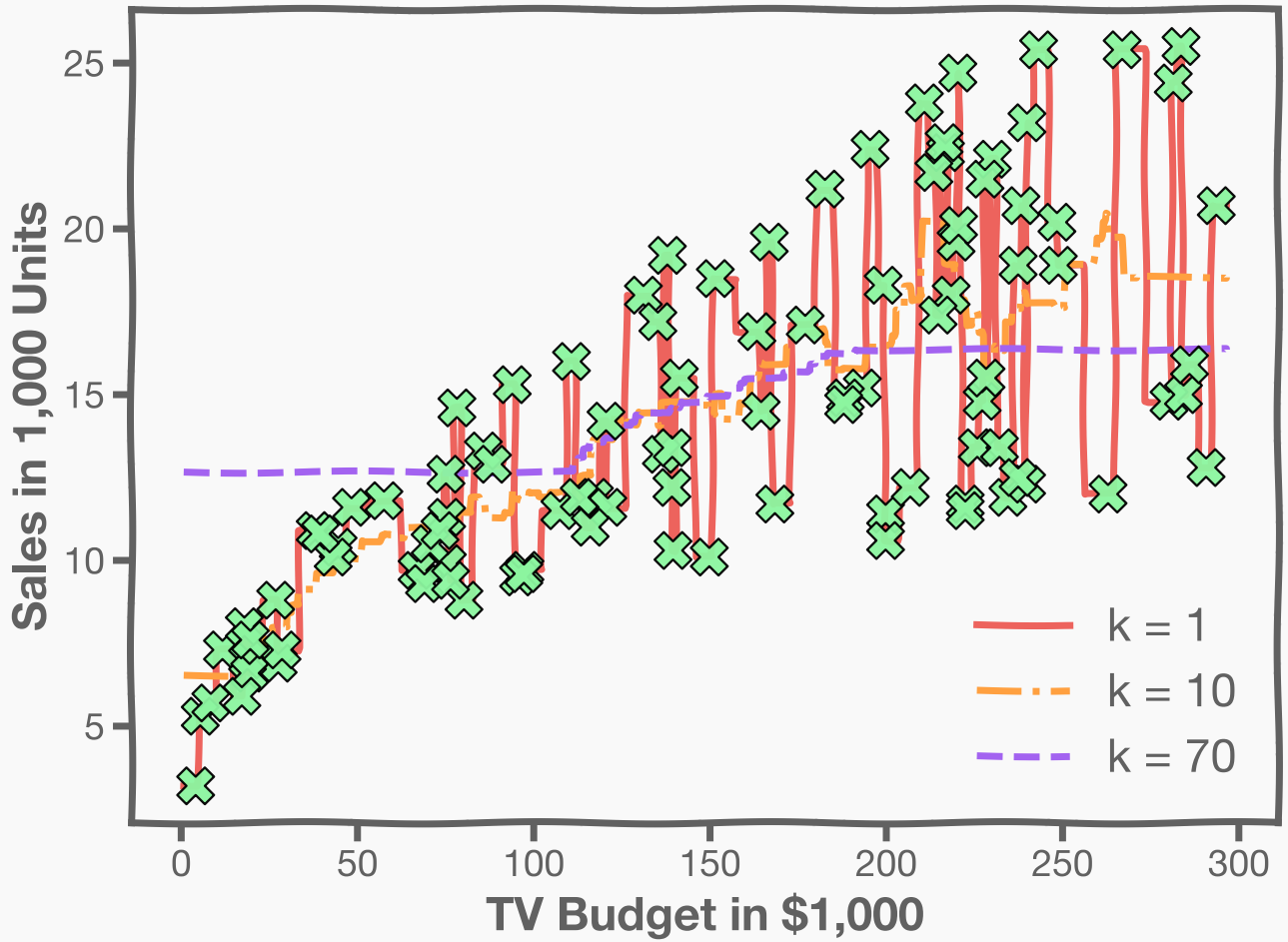
How do we choose from two different models?



https://xkcd.com/833/

k-Nearest Neighbors – kNN

We have tested various models using different k-values on the data.



Choices for model





Which model do you think is the best?

Options:

A. $k = 1$

B. $k = 10$

C. $k = 70$

D. $k = 15$

E. 🤷

Choices for model





Which model do you think is the best?

Options:

$A.\ \ k = 1$

$B.\ \ k = 10$

$C.\ \ k = 70$

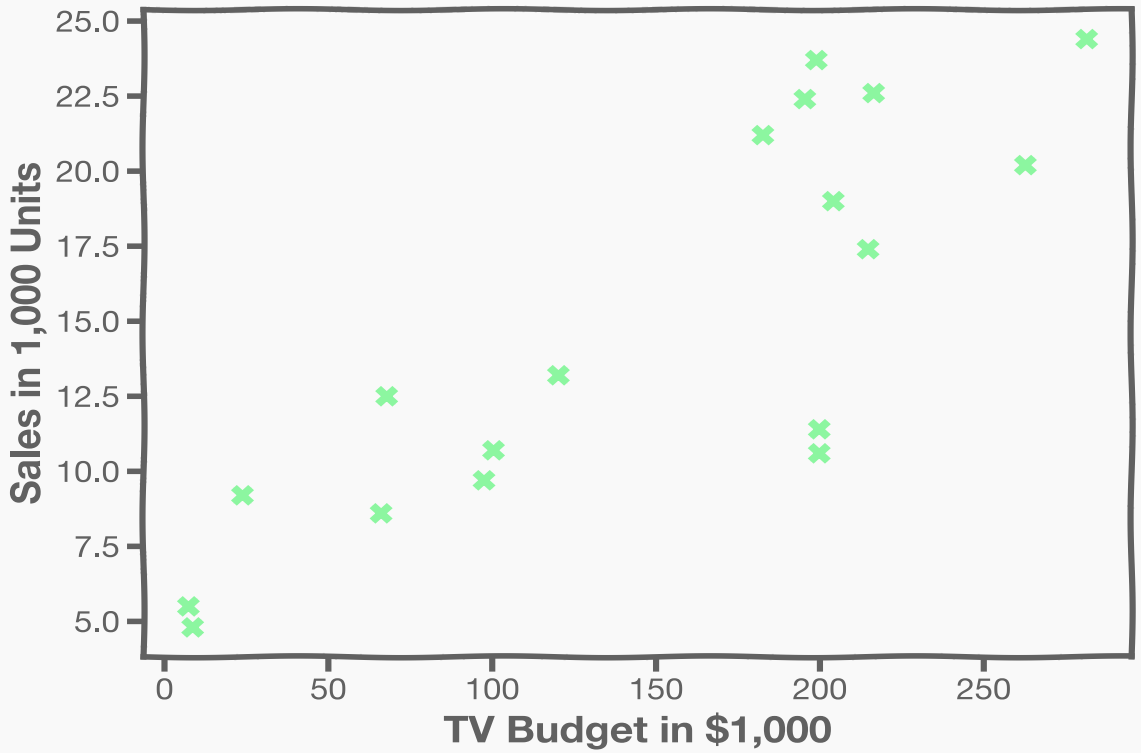
$D.\ \ k = 15$

E. 🤷

Error Evaluation

Error Evaluation

We need to define what we mean by ***best***. To do so, we start with our data.

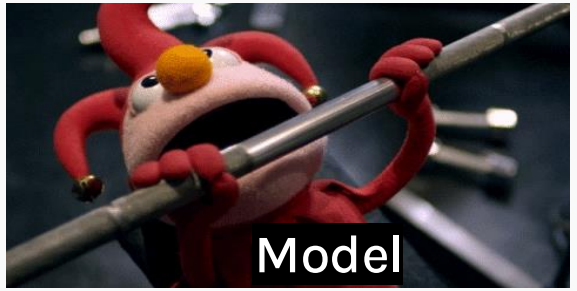


Error Evaluation

We first **withhold** a portion of the data from the model; this process is called **train-validation-test** split.

Train Set

The data that we use to train our model to estimate, $\hat{y}$. x



Validation Set

The data that we use to evaluate our model's performance when choosing different models.



Test Set

The data that we use at the very end to evaluate model's performance on unseen data.



Error Evaluation

The data is randomly shuffled.

We first **withhold** a portion of the data from the model; this process is called **train-validation-test** split.

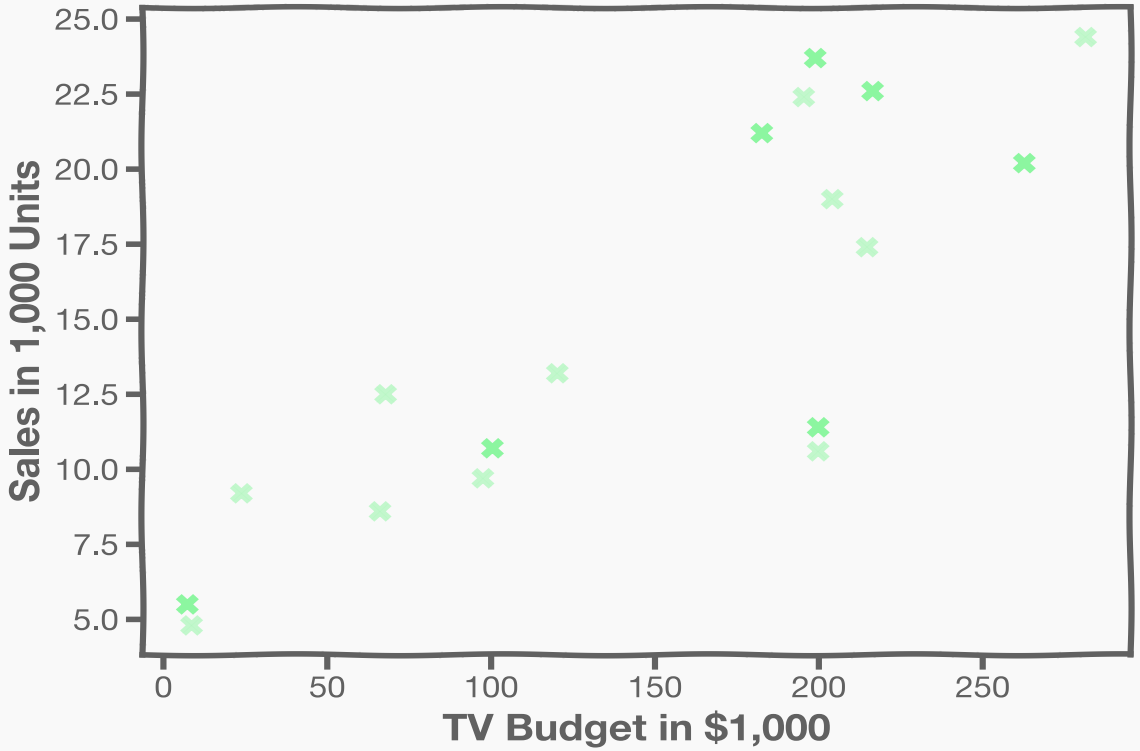
In scikit-learn, we only get a `train_test_split()` function



So, to get a validation set, we split once for train/test, then split the training data again to carve out validation.

Error Evaluation

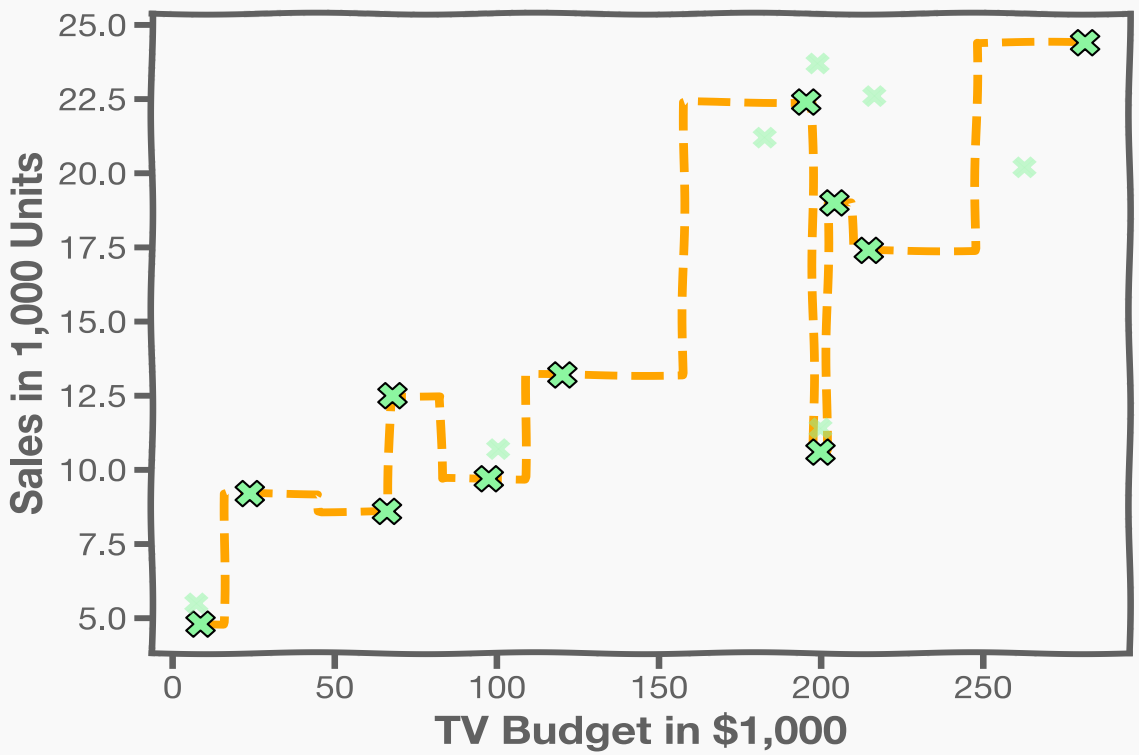
We first **withhold** a portion of the data from the model; this process is called **train-validation-test** split.



We use the training set to estimate $\hat{y}$.

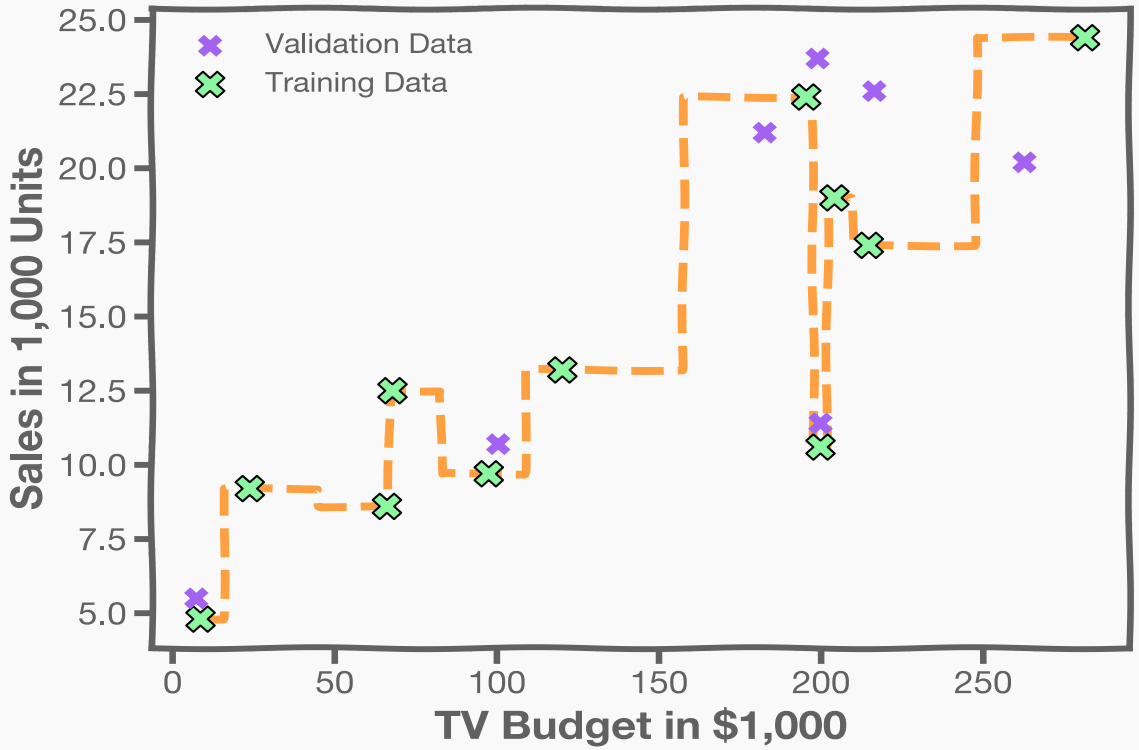
Error Evaluation

Estimate $\hat{y}'s$ values for all the data points in the training set when ***k=1.***



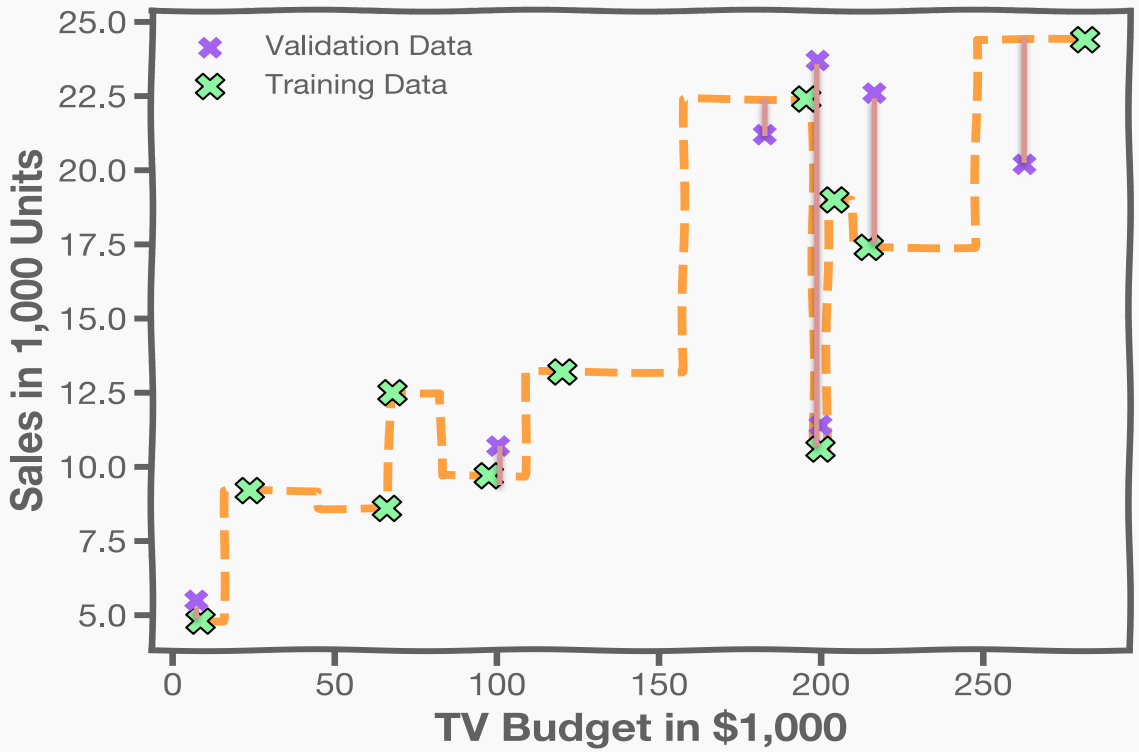
Error Evaluation

Now, we examine the data that was not used for estimating $\hat{y}$, the **validation data** represented by purple crosses.



Error Evaluation

And we calculate the **residuals** $(y_i - \hat{y}_i)$.



For each observation (x_n, y_n), the absolute residuals, $r_i = |y_i - \hat{y}_i|$ quantify the error at each observation point.

Error Evaluation

To quantify the performance of a model, we aggregate the errors. This aggregated value is commonly referred to as the ***loss***, ***error***, or ***cost function***.

A widely used loss function for quantitative outcomes is the **Mean Squared Error (MSE):**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Note: Loss and cost function refer to the same thing. Cost usually refers to the total loss where loss refers to a single training point.

Error Evaluation

Caution: MSE is not the only valid, or necessarily the best, loss function for all scenarios.

Other choices for loss function:

1. Max Absolute Error
2. Mean Absolute Error
3. Huber Loss

We will motivate MSE when we introduce probabilistic modeling.

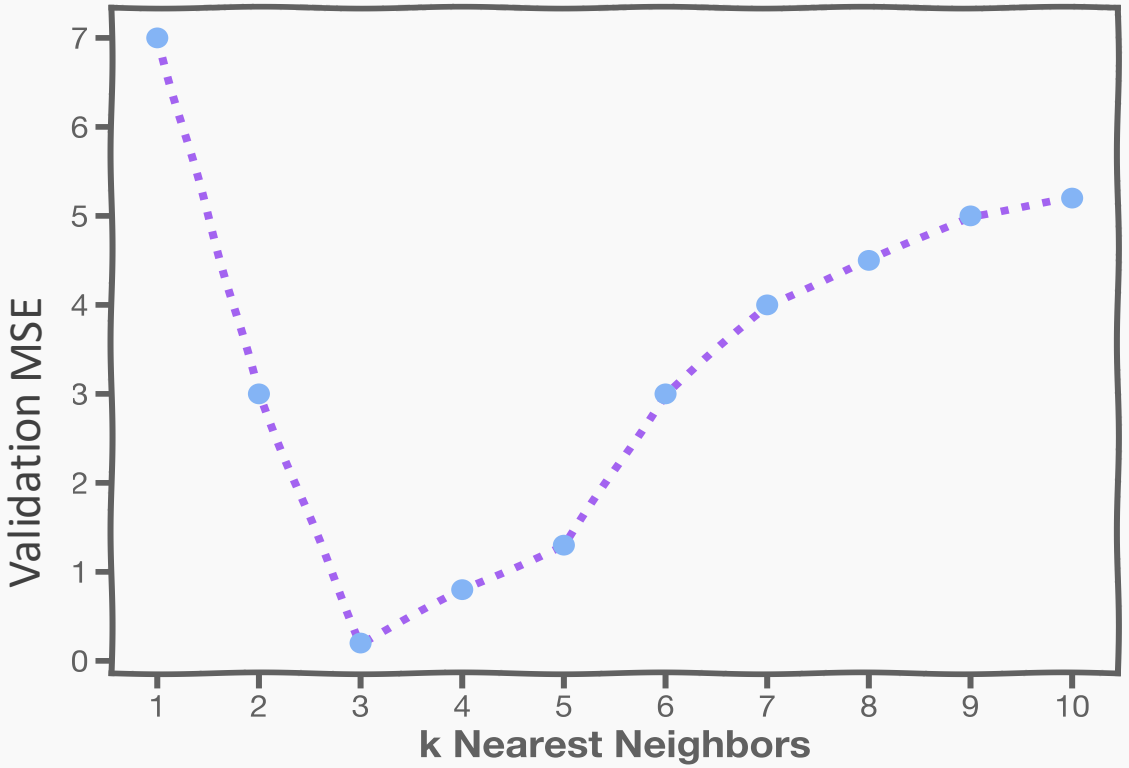
Note: The square **R**oot of the **M**ean of the **S**quared **E**rrors (RMSE) is also commonly used.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Model Comparison

Model Comparison

We repeat this process for all values of k and compare the MSEs on the validation set.



Which model is the best?

Question



Which model do you think is the best now

Options:

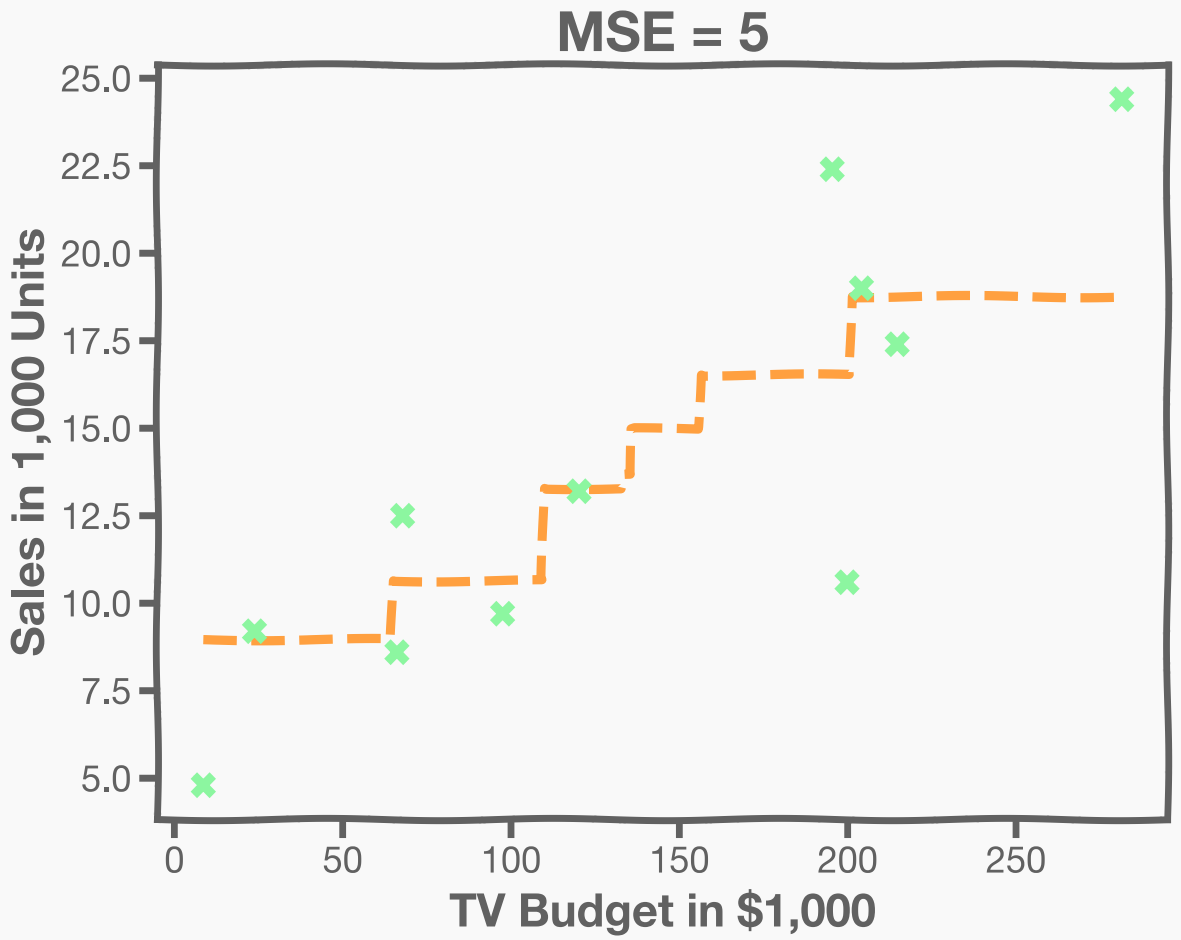
$A.$ $k = 3$

$B.$ $k = 3$ but why don't we experiment with different train/vallidation splits?

Model Fitness

Model fitness

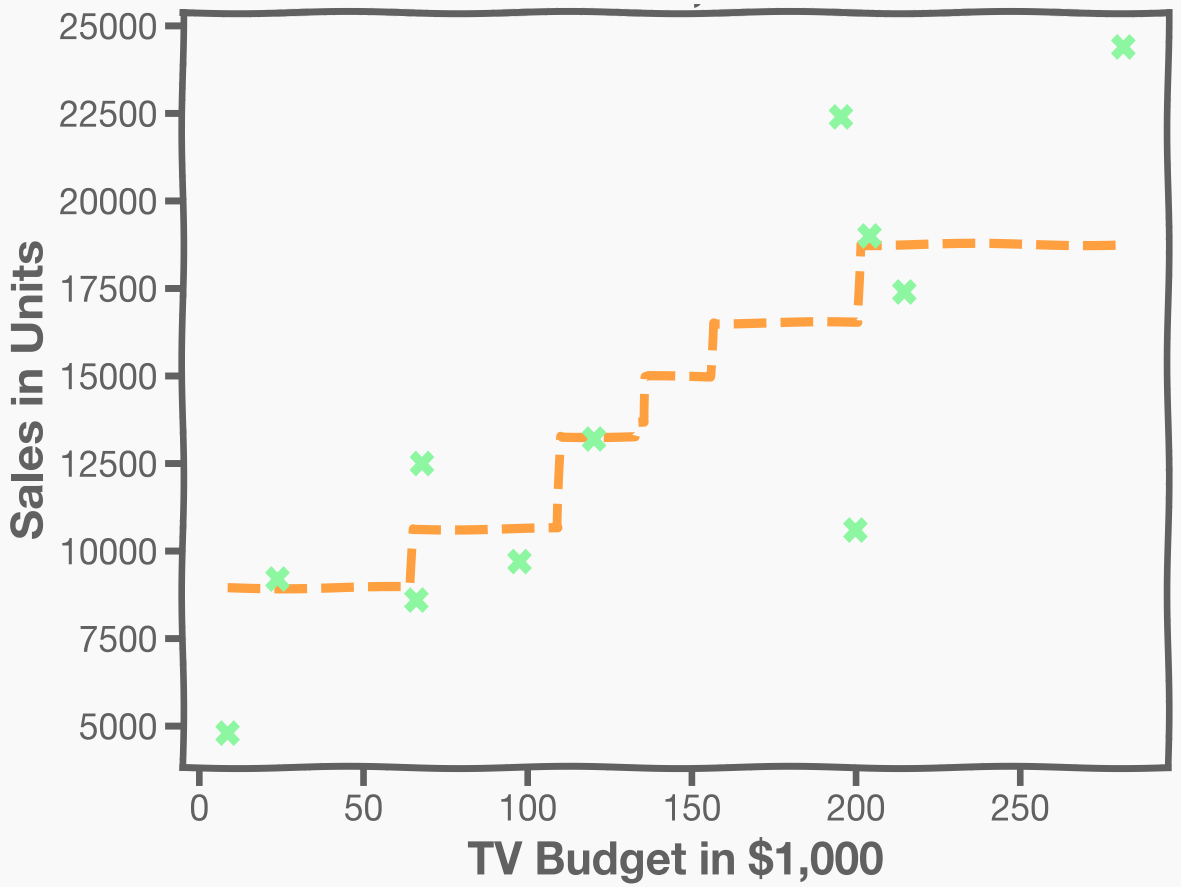
Calculate the MSE for $k = 3$ using a subset of the data.



Is MSE=5.0 good enough?

Model fitness

What would happen if we measure the ***Sales*** in single units instead of 1000 units?



MSE is now 5,004,930.

Is that good?



Model fitness

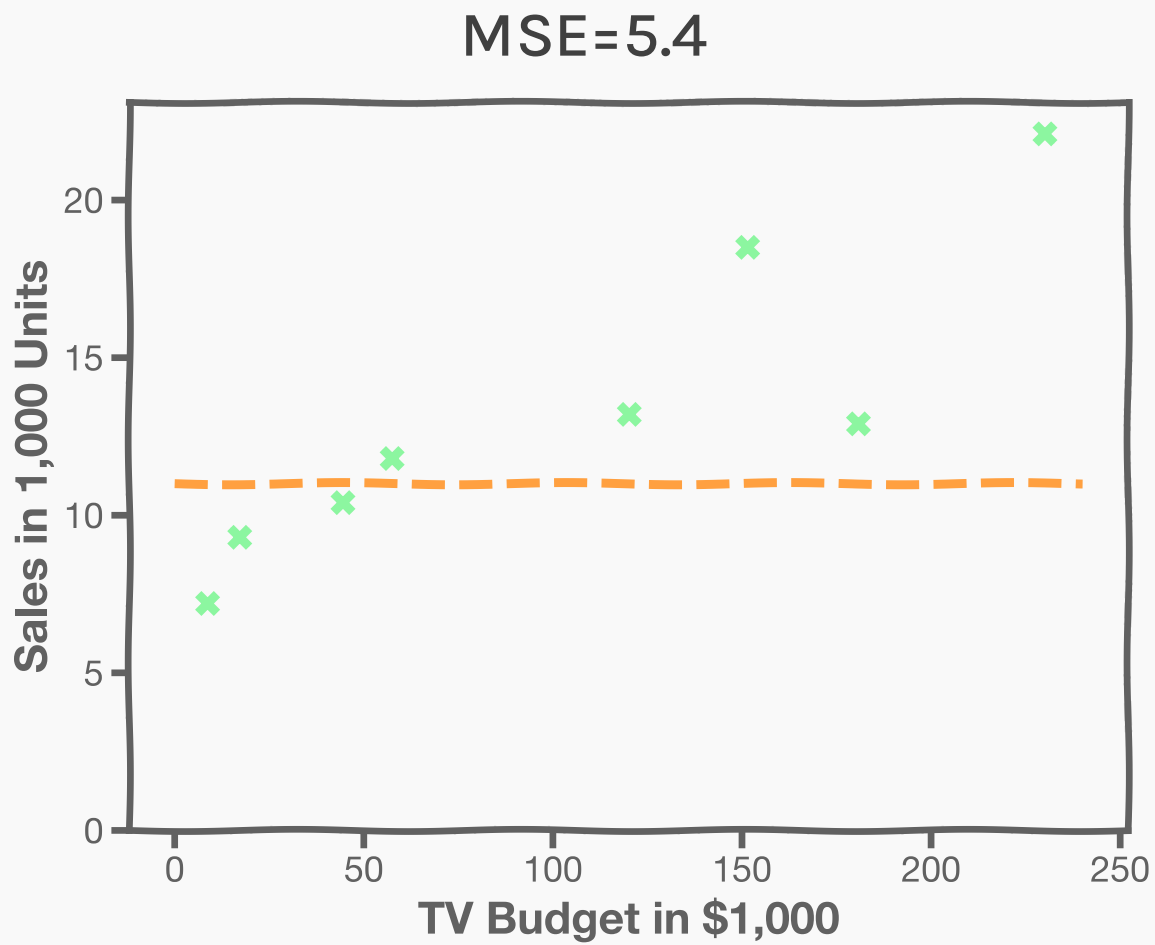
It would be more meaningful to compare it to a benchmark or a known value.

A benchmark that isn't affected by the scale of the data.



Model fitness

It would be more meaningful to compare it to a benchmark or a known value.



We will use the simplest model:

$$\hat{y} = \bar{y} = \frac{1}{n}\sum_i y_i$$

as the worst possible model and

$\hat{y}_i = y_i$ as the best possible model.

R-squared

Though is called R-squared, it is not the square of a quantity R as given in this formula.

We will use two reference models for comparison:

1. The simplest model, often considered the worst possible, where the predicted value $\hat{y}$ is the mean of all observations:

$$\hat{y} = \bar{y} = \frac{1}{n}\sum_i y_i$$

2. The ideal or best possible model, where the predicted value $\hat{y}$ is identical to the actual value y.

Using these two reference models, we define the R^2 (R-squared) value as:

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2}$$

R-squared

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2}$$

- If our model is as good as the mean value, $\bar{y}$, then $R^2 = 0$
- If our model is perfect, then $R^2 = 1$
- R^2 can be negative if the model is worse than the average. This can happen when we evaluate the model in the **validation/test** set.



CS109A
Who Wants to be a Data Scientist?

Consider a dataset with 20 data points. If we perform kNN with k = 20, which of the following statement is true?

Options

A. The R^2 Score will be 0.

B. The MSE will be 0.

C. The model will perform better than if k = 1.

D. The R^2 Score will be very high.

Consider a dataset with 20 data points. If we perform kNN with k = 20, which of the following statement is true?

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A. The R^2 Score will be 0.

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C. The model will perform better than if k = 1.

D. The R^2 Score will be very high.

If your data is noisy and contains a lot of outliers, which of the following options would you consider when using k-NN?

Options

A. Decrease the value of k.

B. Increase the value of k.

C. Leave k as it is and pray that your boss doesn't ask you for a model comparison.

If your data is noisy and contains a lot of outliers, which of the following options would you consider when using k-NN?

Options

A. Decrease the value of k.

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C. Leave k as it is and pray that your boss doesn't ask you for a model comparison.

kNN in Higher Dimensions

- So far, we've looked at kNN in 1D (a single predictor).
- In practice, kNN works with **multiple predictors (features)**.
- Distance is computed in the multi-dimensional space:

$$d\left(\left(x_1, x_2, \dots, x_p\right), \left(x'_1, x'_2, \dots, x'_p\right)\right)$$

Most often: **Euclidean distance**

$$d = \sqrt{\sum_{j=1}^{p} \left(x_j - x'_j\right)^2}$$

The Curse of Dimensionality

- As dimensions increase:
  - Data becomes sparse.
  - "Neighbors" may be far away in high-dimensional space.
- Distances tend to even out → harder to distinguish close vs. far.
- kNN can struggle unless:
  - You have **a lot of data**.
  - You use **feature scaling / selection**.

Takeaway: kNN is intuitive, but high-dimensional spaces make it less effective without care.

Learning Objectives

By the end of this lecture, you should be able to:

- **Define** response and predictor variables, and represent them using the design matrix X and response vector y.
- **Explain** the difference between inference and prediction.
- **Describe** the k-Nearest Neighbors (kNN) algorithm as a **non-parametric method** that relies on data rather than fixed parameters .
- **Implement** kNN in 1D:
    - Find nearest neighbors using distance.
    - Predict outputs by averaging their values.

Learning Objectives

Evaluate kNN model performance using validation/test sets and error metrics.

Extend kNN to multi-dimensional data:

- Compute distances in p-dimensional space (Euclidean distance).
- Recognize scaling and feature importance issues.

Understand the curse of dimensionality and why kNN becomes less effective in high-dimensional spaces.


TRENDIZISST
THANK YOU!!